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In quantum mechanics, particles tunnel through classically forbidden regions ($E < V(x)$), so we define bound and scattering states based on asymptotic potential limits:

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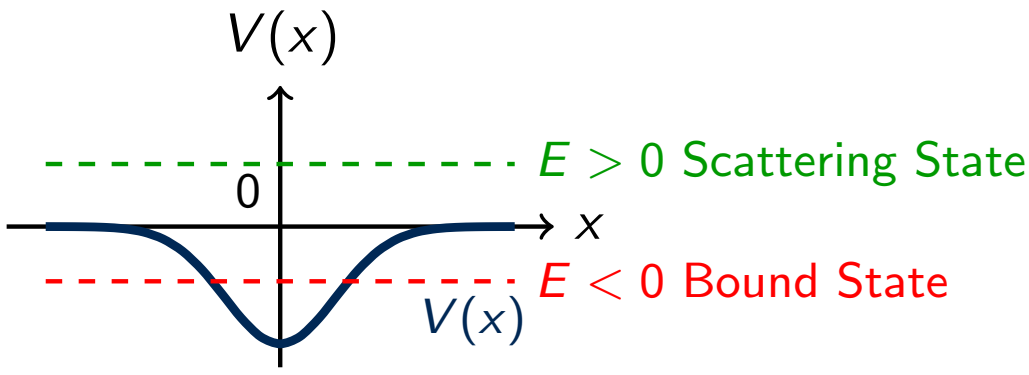
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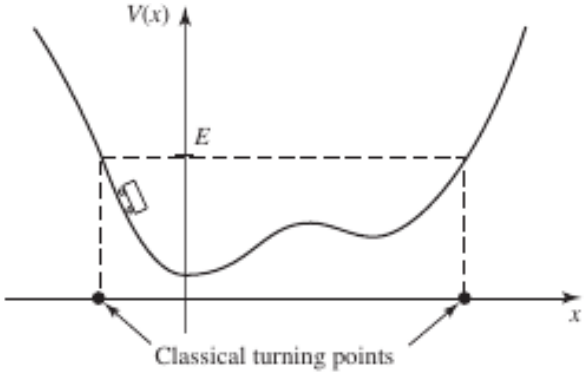
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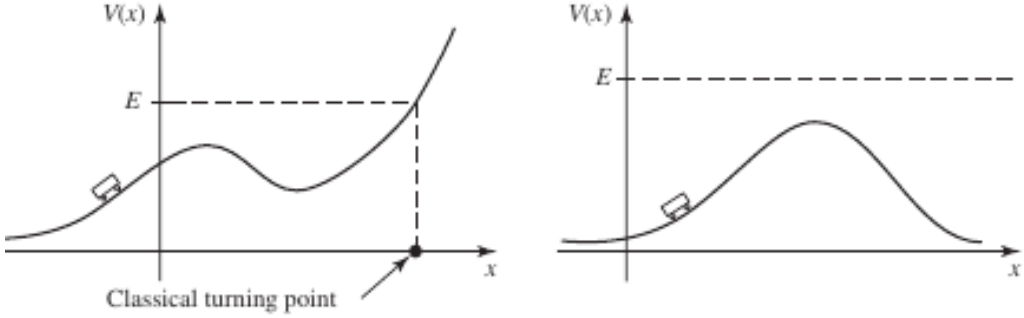
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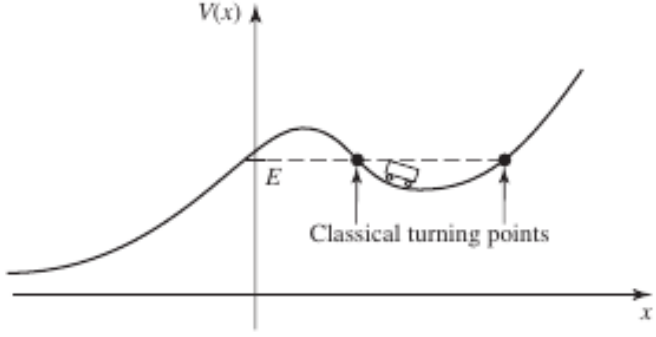
(a)

bound state : discrete spectrum



(b)

Scattering states : continuous spectrum



(c)

classical bound state
but quantum scattering state

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3. No Degeneracy in 1D Bound States

A fundamental theorem in 1D quantum mechanics guarantees that there are no degenerate bound states for a given energy E.

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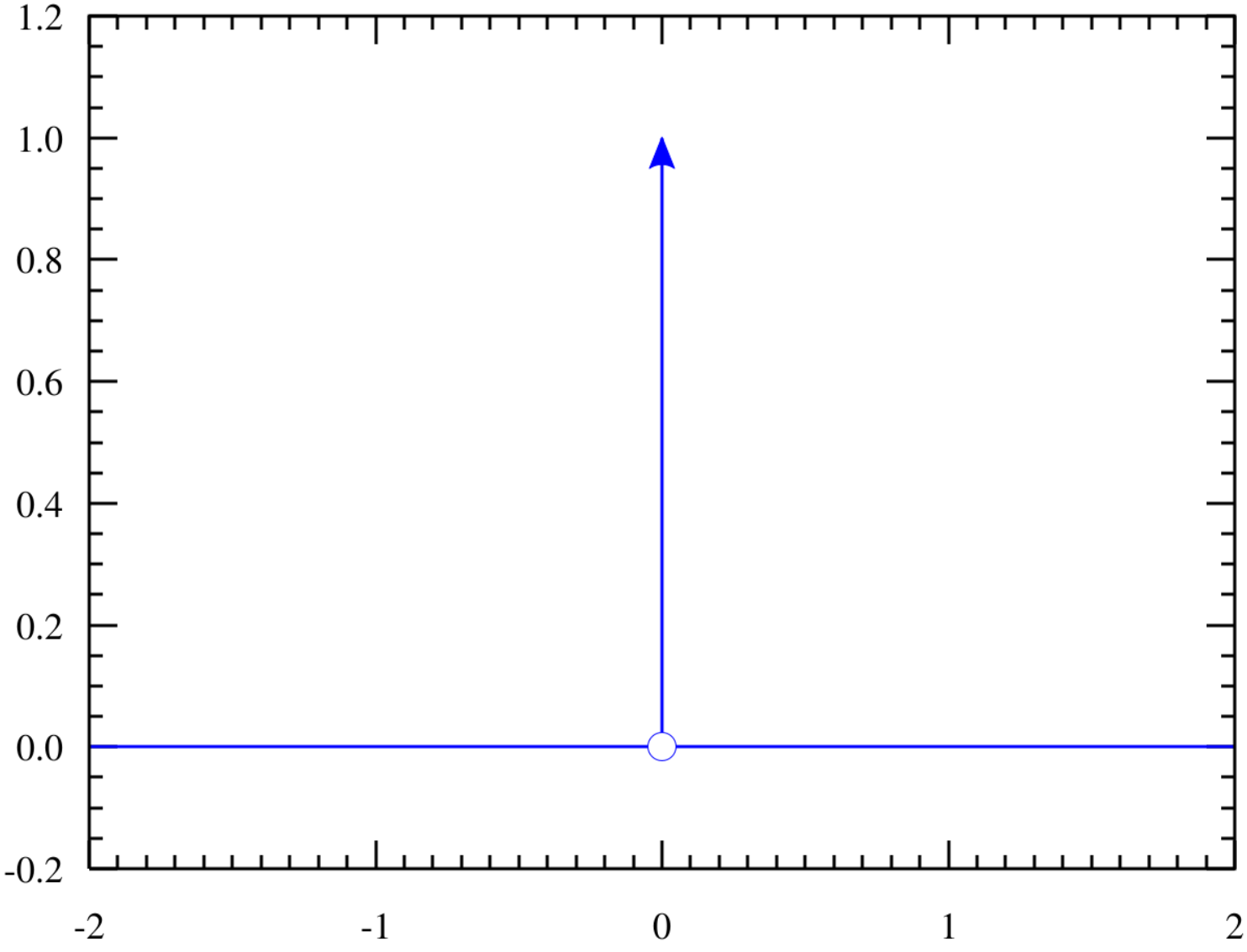
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continuum basis of infinite dimension : operators have continuum spectrum and eigen values are non-normalizable, $\hat{A}|k\rangle = k|k\rangle$; $\langle k'|k\rangle = \delta(k' - k)$
and $\delta(x)$ is defined as 0 everywhere else except at $x = 0$,
s.t. $\int_{-\infty}^{\infty} \delta(x) = 1 \Leftarrow$ Dirac delta function



Height of the arrow usually specifies value of multiplicative constant,

which gives the area under the function

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2. Normalization via Wave Packets : Individual energy eigenfunctions $\psi_E(x)$ are non-normalizable: $\int_{-\infty}^{\infty} |\psi_E(x)|^2 dx = \infty$
Physical states are constructed as superpositions (wave packets):

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int \phi(k)\psi_k(x)e^{-iE_k t/\hbar}\, dk$$

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Physical Meaning	Trapped particle	Incident /Reflected /Transmitted

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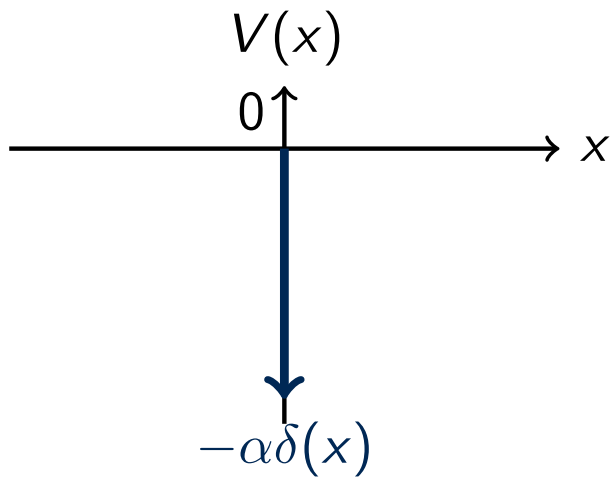
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- Area under the spike is normalized:

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1$$



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Integrating from $-\epsilon$ to $+\epsilon$ and taking the limit $\epsilon \to 0$:

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$\frac{d\psi}{dx}$ undergoes a finite jump discontinuity

$\propto$ strength α of the delta function at $x = 0$

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Energy Eigenvalue : Substituting $\kappa = \frac{\sqrt{-2mE}}{\hbar} \implies$ exactly one bound state:

$$E = -\frac{\hbar^2\kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}$$

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$$\int_{-\infty}^{\infty} |A|^2 e^{-2\kappa|x|} dx = 2|A|^2 \int_0^{\infty} e^{-2\kappa x} dx = 2|A|^2 \left[\frac{1}{2\kappa}\right] = \frac{|A|^2}{\kappa} = 1$$

Choosing A to be real and positive:

$$A = \sqrt{\kappa} = \frac{\sqrt{m\alpha}}{\hbar}$$

Complete Bound State Solution

$$\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-\frac{m\alpha}{\hbar^2}|x|}, \qquad E = -\frac{m\alpha^2}{2\hbar^2}$$

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Assuming particles are incident from the left only, we set $G = 0$:

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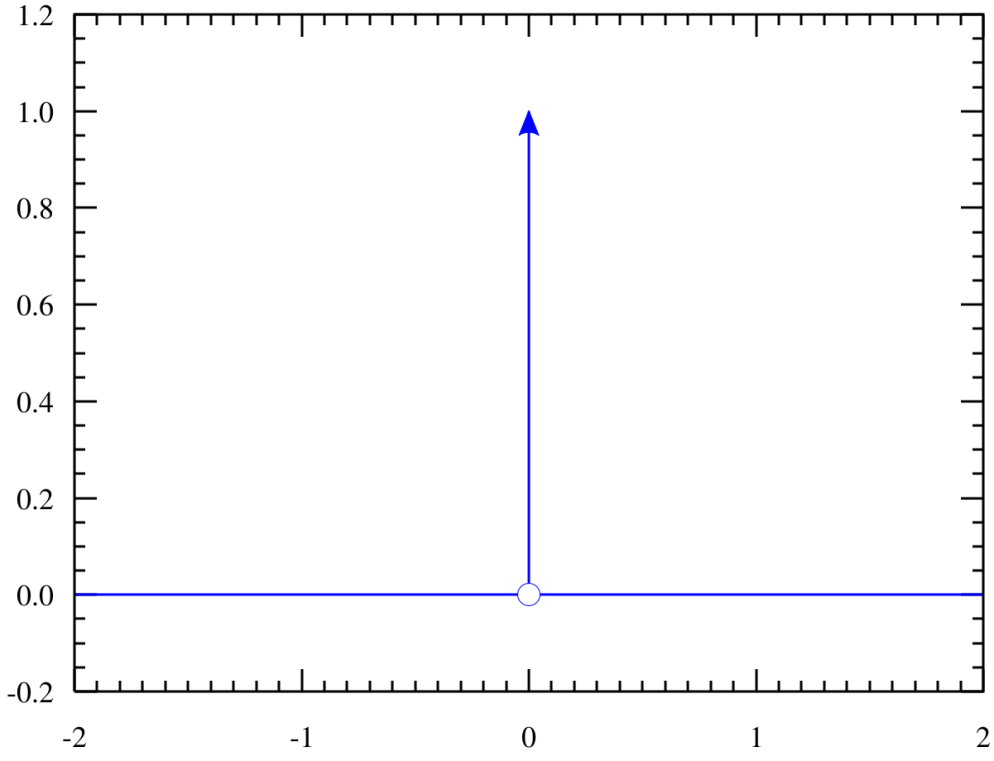
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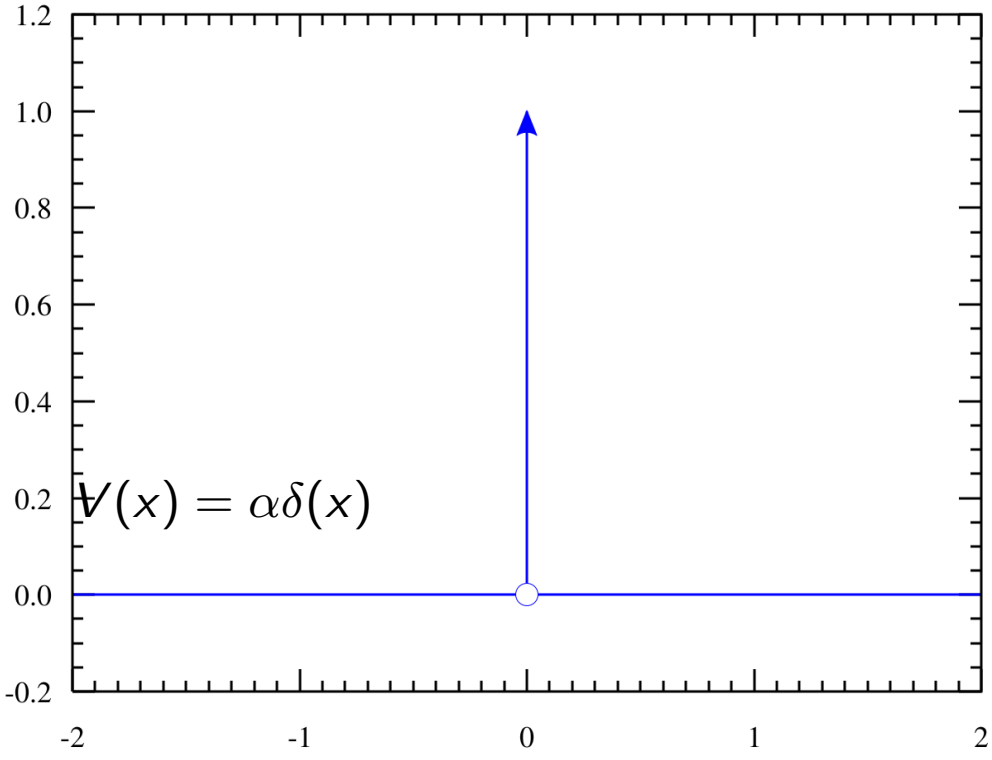
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Conservation of Probability : $R + T = \frac{\beta^2}{1+\beta^2} + \frac{1}{1+\beta^2} = 1 \implies$, conservation of probability flux across the delta-function potential well.

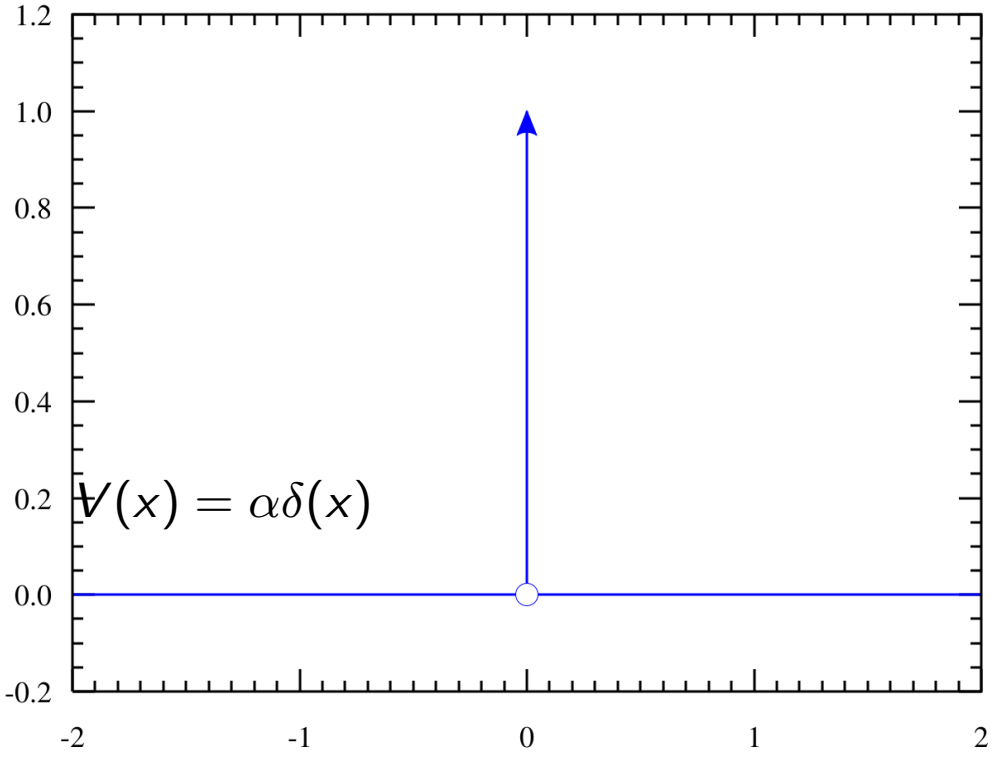
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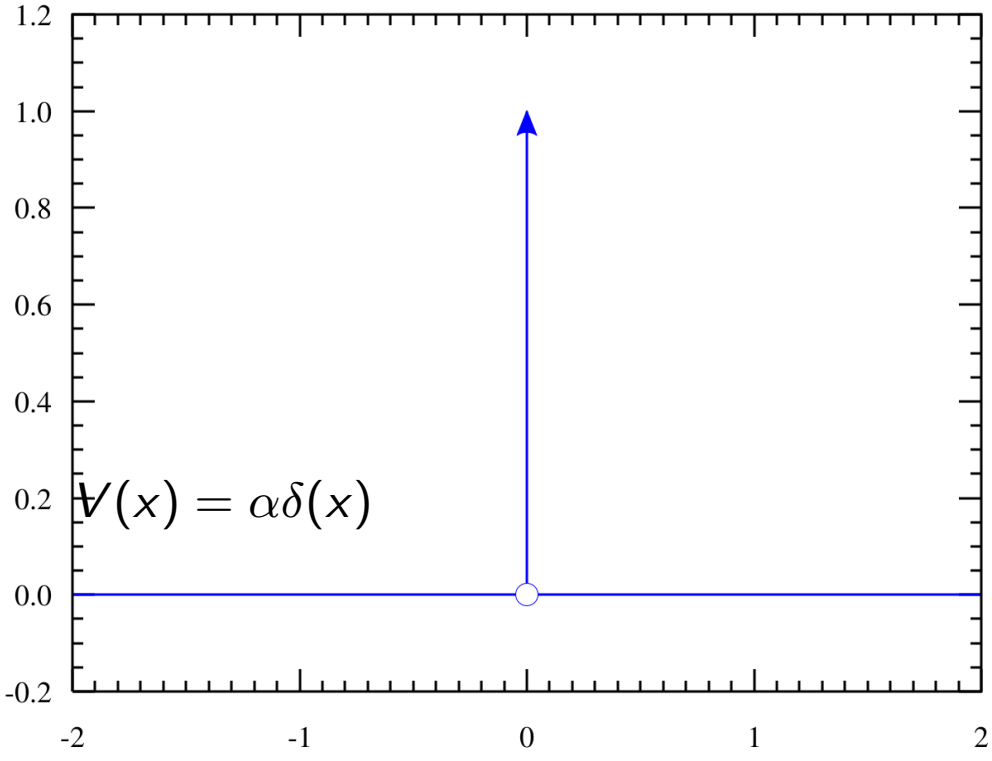


delta-function barrier



No bound state

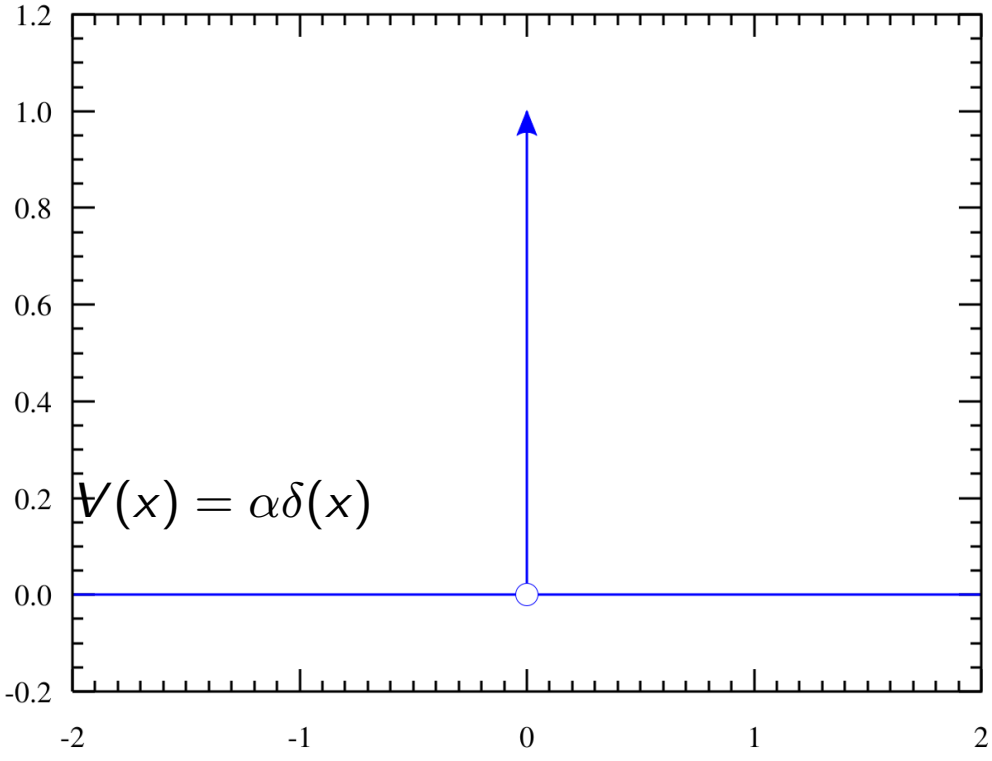
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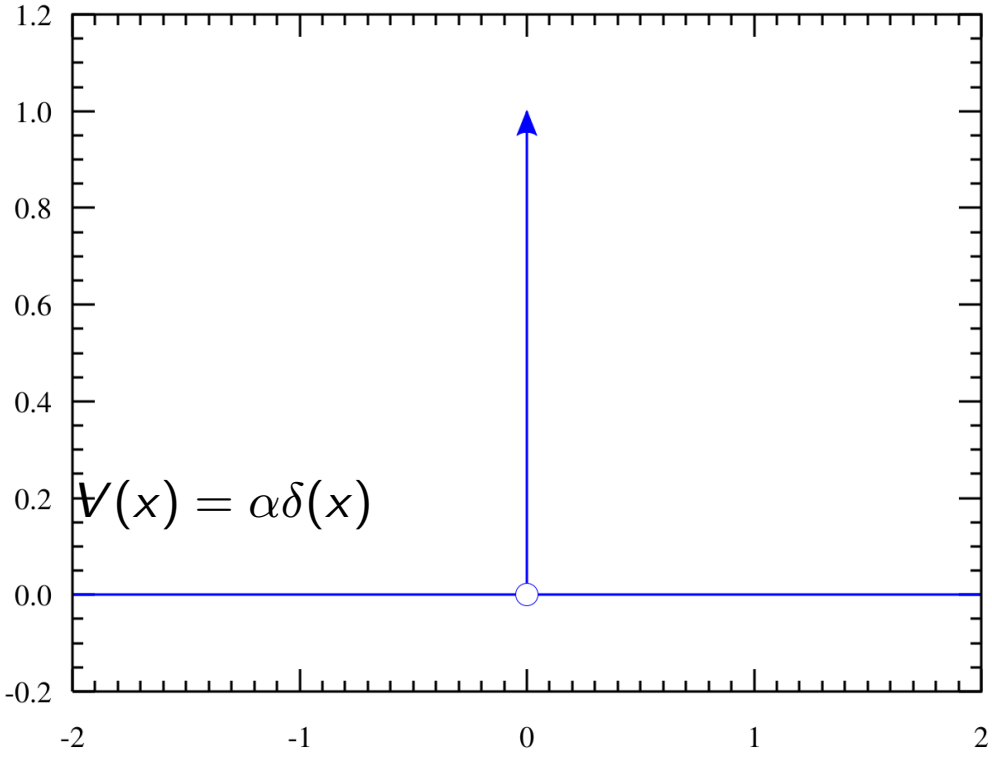


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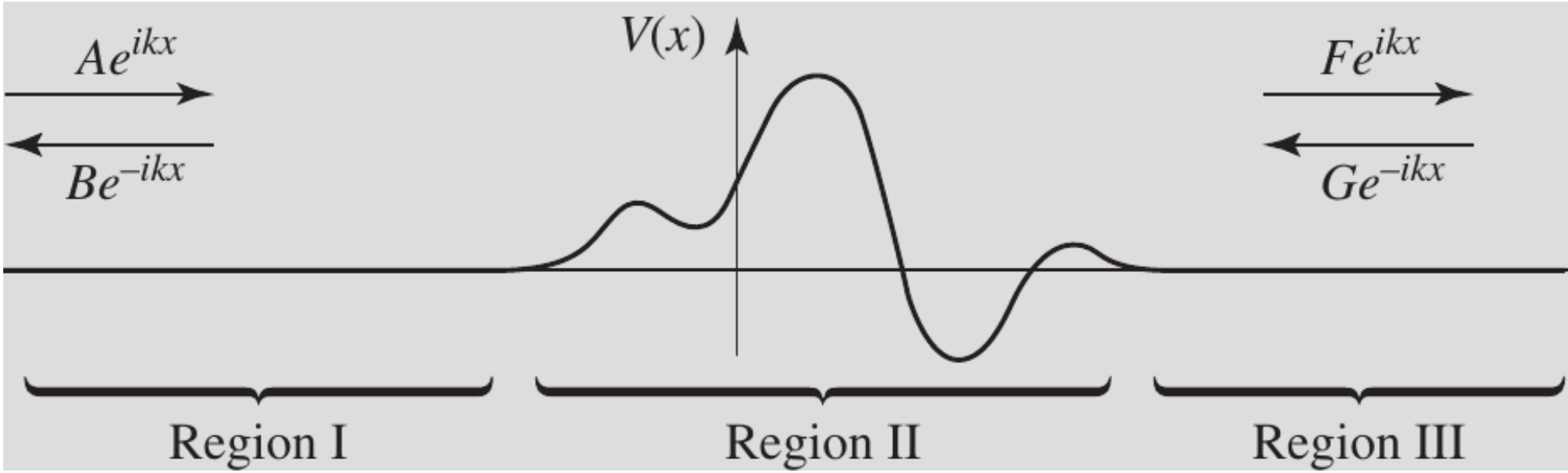
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Particle just as likely to pass through the barrier as to cross over the well!

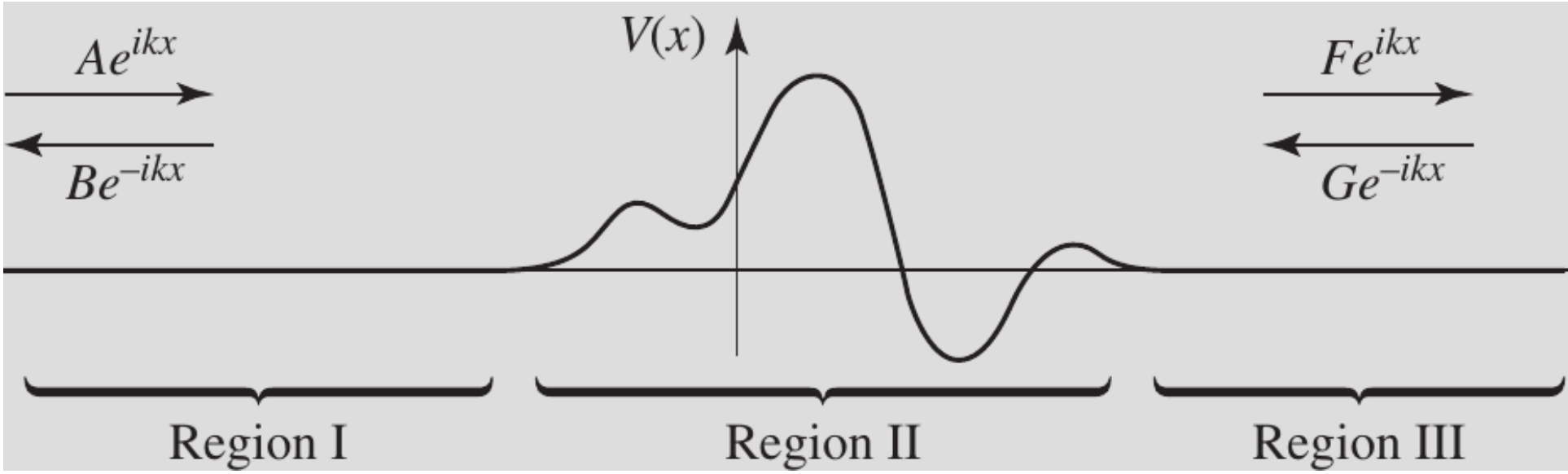
The scattering matrix

Consider an arbitrary localized potential



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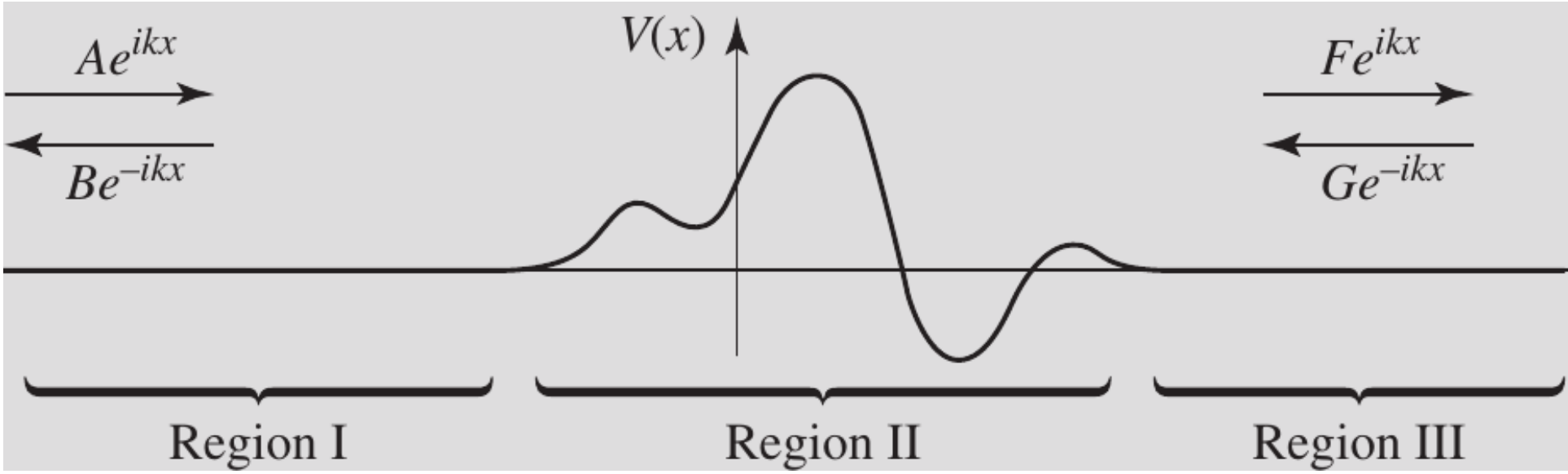
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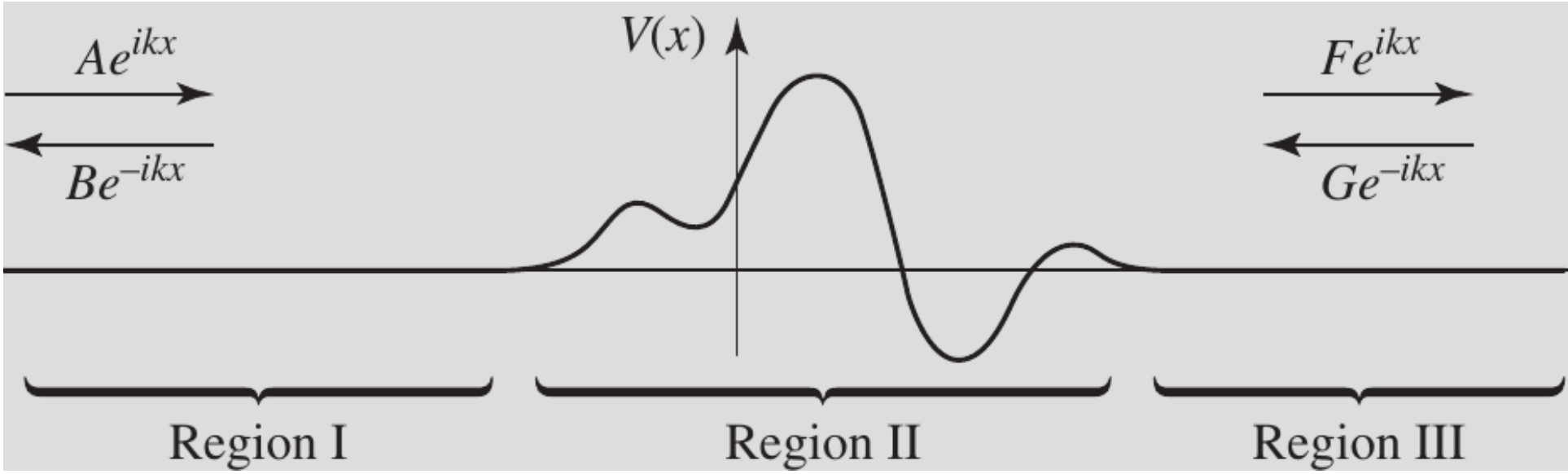


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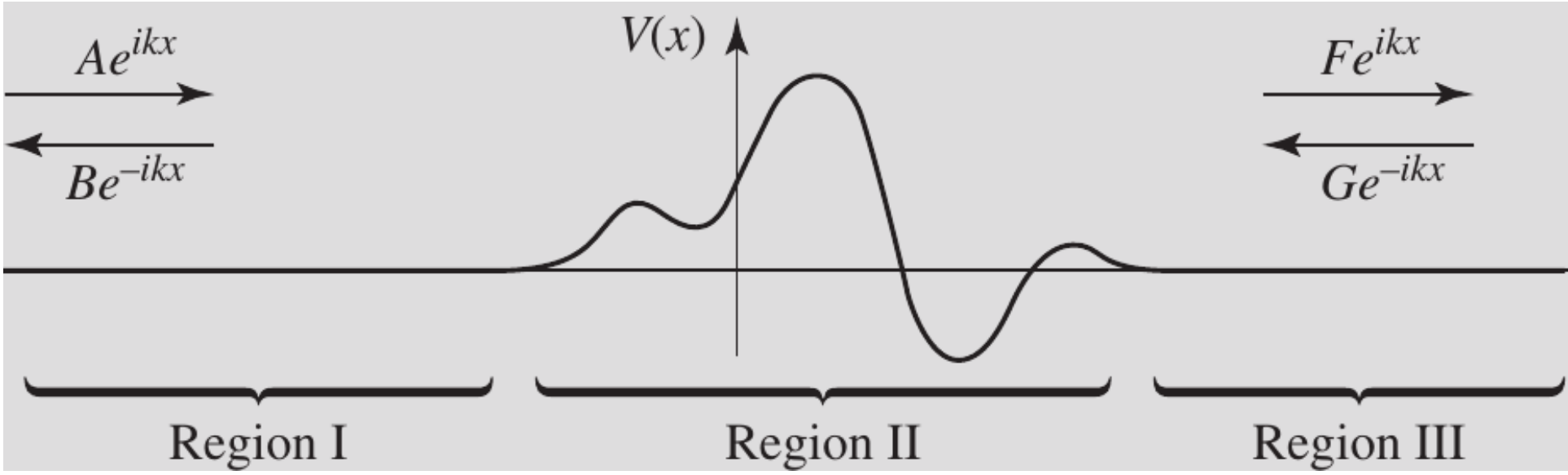
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where $f(x)$ and $g(x)$ are two linearly independent particular solutions

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The relation between the incoming wave amplitudes (A, F) and the outgoing wave amplitudes (B, G) for a scattering potential is linear

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$$\Rightarrow S^\dagger S = I$$

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